

SNIGDHA GAYATHRI MANDAPATI

AI Engineer | LLM Systems, RAG Pipelines & Multi-Agent Orchestration

+91 8247310334 | snigdhaisme@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

AI Engineer with a strong foundation in machine learning, deep learning, and algorithm design, applying Python, TensorFlow, and PyTorch across the full model lifecycle: **architecture, training, evaluation (F1, ROC-AUC, RMSE)**, and production deployment. Track record of profiling and **eliminating performance bottlenecks in deployed ML systems**, cutting inference latency and processing time through pipeline re-engineering, caching, and precomputation. Builds on **linear-algebra-grounded techniques** (embedding spaces, similarity kernels, dimensionality reduction) in **retrieval and recommendation systems**. Strong communicator and technical writer, documenting architecture and benchmark results for production RAG and multi-agent systems shipped to end users.

SKILLS

Programming & Algorithms: Python, Java, C++, JavaScript/TypeScript, SQL, algorithm design, software design, data structures

ML & Deep Learning: TensorFlow, PyTorch, Scikit-Learn, XGBoost, Deep Learning, NLP, model evaluation, linear algebra

Performance & Systems: Latency optimization, pipeline profiling, model monitoring, drift detection, FastAPI, Flask REST APIs, Docker, CI/CD

LLM & Generative AI: Large Language Models, Transformer Architecture, Context Engineering, Prompt Engineering, LLM Evaluation & Benchmarking, LLMops, Hugging Face Transformers, Gemini API

Agentic AI & Orchestration: LangChain, LangGraph, LlamaIndex, Multi-Agent Systems, Tool Use / Function Calling, MCP

RAG & Retrieval: RAG, Vector Databases, Embedding Models (SentenceTransformers), Semantic Search, Reranking, RAGAS

Fine-Tuning: LoRA / QLoRA, Supervised Fine-Tuning (SFT), Quantum ML (PennyLane)

Data & Infra: PostgreSQL, MySQL, Next.js, React, Git, JWT/Auth (Clerk)

TECHNICAL EXPERIENCE

Machine Learning Intern | Cognifyz Technologies

Jan 2025 – Mar 2025

- Sustained **sub-200ms latency** across **10K+ daily production requests** by deploying **10+ supervised ML models** (Python, Scikit-Learn, XGBoost) as Dockerized Flask REST APIs.
- Reduced production **model drift by ~30%** and achieved **95% validation accuracy** on skewed real-world datasets by designing and shipping automated monitoring and retraining pipelines.

Artificial Intelligence Intern | Technical Hub Pvt. Ltd.

May 2024 – Sep 2024

- Increased **retrieval precision by 40% over the prior baseline** by architecting **multi-stage classification pipelines** (PCA, ensemble methods) with iterative, statistically-grounded feature engineering.
- Cut inference **pipeline processing time by 35%** and accelerated **model iteration cycles by 25%** by profiling and re-engineering the ML pipeline algorithm-by-algorithm.

PROJECTS

Agentic Placement RAG | Security-Hardened, Production-Grade RAG Pipeline

[Link](#)

- Delivered a security-hardened Agentic RAG platform, achieving a **9-stage guardrail pipeline** and **≥90% test coverage**, by architecting a **FastAPI + LangGraph (Python) backend** with **hybrid BM25 + Gemini retrieval**.
- Improved **retrieval precision** and **reduced hallucinations** in a live RAG assistant, verified via **cross-encoder re-ranking** and **grounded-output checks**, by building **multi-hop query rewriting** and shipping to production on Render.

Smart Shelf AI | Hybrid Quantum-Classical Recommendation Engine

[Link](#)

- Cut recommendation latency** from a 1–2 min cold start to **sub-1-second on repeat queries**, an 80x+ improvement, by engineering a hybrid **classical-quantum pipeline** (SentenceTransformers, PennyLane 6-qubit kernel) with precomputed caching.
- Delivered a fully private recommendation engine with **100% local inference** and **zero external dependencies** by building **27-class emotion detection** (GoEmotions, PyTorch/TensorFlow) on a dedicated **FastAPI backend**.

Enterprise Knowledge Intelligence System | Agentic Enterprise Knowledge Platform with Hybrid Retrieval

[Link](#)

- Engineered EKIP using **LangGraph, FastAPI, Qdrant, Neo4j**, and Gemini, enabling enterprise-scale knowledge discovery with **4-agent orchestration** and hybrid **vector-graph retrieval**, reducing complex query resolution time by **35%**.
- Designed a **hybrid RAG pipeline** combining semantic search with **knowledge graph traversal**, improving **retrieval accuracy by 22%** and enabling explainable multi-hop reasoning across enterprise knowledge.

EDUCATION

B.Tech in Computer Science & Engineering (AI/ML) — Aditya University, Surampalem

2022 – 2026 | CGPA: 8.30

CERTIFICATIONS

Microsoft (**Azure AI Fundamentals**) | Google (Career Essentials in **Generative AI**) | IBM (**ML with Python, DL Fundamentals**, Data Science 101) | LangChain Academy (Introduction to **LangChain** – Python, Building **Reliable Agents**) | SimpliLearn (**LangChain for LLM App Development**) | ActiveLoop (**Retrieval Augmented Generation for Production** with LangChain & LlamaIndex) | Pearson (IT Specialist – **Java**)